ENGINEERING ECONOMICS (CE 655)

Chapter 3: Basic Methodologies of Engineering Economic Analysis

Questions & Answers — TU Past Exam Papers (2070–2082)

CHAPTER 3 SYLLABUS OVERVIEW (6 Hours)

Topics:

(a) MARR — Definition, Factors affecting MARR

(b) Payback Period — Standard and Discounted

(c) Equivalent Worth Methods — Present Worth (PW), Future Worth (FW), Annual Worth (AW)

(d) Rate of Return Methods — IRR and ERR

(e) Benefit-Cost Ratio — Conventional and Modified

(f) Financial vs Economic Analysis, Life Cycle Costing

SECTION A: MARR (Minimum Attractive Rate of Return)

Q1. Define MARR. What are the factors affecting MARR? [2081 Ashwin Q3a, 2072 Magh Q3a — 3 Marks]

Answer:

Definition of MARR

The Minimum Attractive Rate of Return (MARR) — also called the Hurdle Rate — is the minimum interest rate that an organisation requires to earn on any investment before it will be accepted. A project is acceptable only if its IRR ≥ MARR, or equivalently, NPW ≥ 0 when discounted at MARR.

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Decision Rule:  If IRR ≥ MARR  →  ACCEPT  |  If IRR < MARR  →  REJECT
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Factors Affecting MARR:

1. Cost of Capital: The weighted average cost of debt and equity financing. MARR must at least cover the cost of acquiring capital.
2. Opportunity Cost: Return available on the best foregone alternative. MARR should not be lower than the best available alternative return.
3. Risk Level: Higher project risk (technical, market, political) requires a higher MARR to compensate for uncertainty.
4. Type of Organisation: Public sector uses lower MARR (social discount rate ~6–12%); private sector uses higher MARR (~15–25%).
5. Availability of Capital: When investment funds are scarce and many projects compete, MARR is raised to ration capital.
6. Tax Considerations: Income taxes reduce after-tax returns; before-tax MARR must be set higher to achieve required after-tax return.
7. Inflation Rate: High inflation requires a higher nominal MARR to maintain a satisfactory real rate of return.
8. Liquidity Needs: Organisations needing ready cash set higher MARR to filter out long-term, illiquid investments.

9. Government Policy: Interest rate policies, subsidies, and regulations affect the prevailing cost of capital.
10. Competition and Market Conditions: High-competition or high-technology markets raise MARR due to shorter product cycles and higher risk.

Factor	Effect on MARR
High project risk	Increases MARR
Scarce capital / many projects	Increases MARR
Government / public project	Decreases MARR
High inflation	Increases MARR
Low cost of borrowing	Decreases MARR

SECTION B: Payback Period Method

Q2. Write advantages and limitations of payback period method with suitable example. [2080 Chaitra Q3a — 4 Marks]

Answer:

Payback Period Method

The Payback Period (PBP) is the length of time required to recover the initial investment from the net cash flows generated by a project, without considering the time value of money.

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PBP = Initial Investment / Annual Net Cash Flow   (for uniform cash flows)
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Advantages:

1. Simplicity: Very easy to calculate and interpret, requiring no estimation of MARR.
2. Liquidity Focus: Directly shows how quickly the initial investment is recovered — important for cash-constrained firms.
3. Risk Indicator: Shorter PBP means less exposure to uncertain future cash flows.
4. Useful Screening Tool: Quickly filters out obviously poor investments before detailed analysis.

Limitations:

1. Ignores Time Value of Money: Rs. 10,000 received in year 1 treated the same as Rs. 10,000 in year 5 (corrected by Discounted PBP).
2. Ignores Post-Payback Cash Flows: Cash flows after payback are completely ignored, penalising long-lived profitable projects.
3. No Profitability Measure: Does not reflect total return or NPW of the investment.
4. Arbitrary Cutoff Period: The acceptable payback benchmark is set subjectively with no theoretical basis.
5. Cannot Rank Mutually Exclusive Projects Correctly: May prefer a project with a shorter PBP but lower NPW.

Example:

Project A: Investment Rs. 60,000, annual net cash flow Rs. 20,000 for 5 years. PBP = 60,000/20,000 = 3 years. Total return = Rs. 1,00,000.

Project B: Investment Rs. 60,000, cash flows: Rs. 50,000 (yr1), Rs. 10,000 (yr2), then Rs. 5,000/year for years 3–10. PBP = 2 years. Total return = Rs. 95,000.

PBP ranks B better (2 years < 3 years), but B earns less total value. This illustrates the limitation of ignoring post-payback cash flows.

Q3. Assess project feasibility using Discounted Payback Period, AW, IRR, and ERR. MARR = 15%, ε = 18%. [2080 Chaitra Q3b — 2+2+2+2 Marks]

Answer:

Cash Flow (Rs. '000):

EOY	0	1	2	3	4	5
CF (Rs.'000)	-12	7	-6	14	5	2

(i) Discounted Payback Period at MARR = 15%:

EOY	CF (Rs.'000)	PW Factor @15%	PW of CF	Cumulative PW
0	-12	1.0000	-12.000	-12.000
1	7	0.8696	6.087	-5.913
2	-6	0.7561	-4.537	-10.450
3	14	0.6575	9.205	-1.245
4	5	0.5718	2.859	1.614
5	2	0.4972	0.994	2.608

DPBP = 3 + 1.245/2.859 = 3.44 years (≈ 3 years 5 months)

DPBP < project life (5 years): ACCEPTABLE.

(ii) Annual Worth (AW) at MARR = 15%:

PW_total = -12 + 6.087 - 4.537 + 9.205 + 2.859 + 0.994 = +2.608 (Rs. 2,608)

AW = 2.608 × (A/P,15%,5) = 2.608 × 0.2983 = Rs. 778

AW = Rs. 778 > 0: ACCEPTABLE.

(iii) IRR — PW = 0:

Try i=21%: PW = -12+7(0.8264)-6(0.6830)+14(0.5645)+5(0.4665)+2(0.3855)
= -12+5.785-4.098+7.903+2.333+0.771 = +0.694

Try i=25%: PW = -12+7(0.8000)-6(0.6400)+14(0.5120)+5(0.4096)+2(0.3277)
= -12+5.600-3.840+7.168+2.048+0.655 = -0.369

IRR = 21 + 4×0.694/(0.694+0.369) = 21+2.61 = 23.61%

IRR = 23.61% > MARR (15%): ACCEPTABLE.

(iv) ERR — ε = 18%, MARR = 15%:

PW_neg = -12 + (-6)(P/F,15%,2) = -12 - 4.537 = -16.537 (Rs. '000)

FW_pos = 7(F/P,18%,4)+14(F/P,18%,2)+5(F/P,18%,1)+2

= 7(1.9388)+14(1.3924)+5(1.18)+2 = 13.572+19.494+5.900+2 = 40.966

16.537(1+ERR)^5 = 40.966 => (1+ERR)^5 = 2.4773

$$ERR = 2.4773^{(0.2)} - 1 = 1.1993 - 1 = 19.93\%$$

ERR = 19.93% > MARR (15%): ACCEPTABLE. All four methods confirm feasibility.

Q4. Evaluate the Discounted Payback Period at MARR = 10%. Standard PBP = 4 years. [2078 Baishakh Q6 — 4 Marks]

Answer:

EOY	Net CF (Rs.)	PW Factor @10%	PW of CF	Cumulative PW
0	-10,000	1.000	-10,000	-10,000
1	2,000	0.9091	1,818	-8,182
2	3,100	0.8264	2,562	-5,620
3	4,200	0.7513	3,155	-2,465
4	5,300	0.6830	3,620	1,155
5	6,400	0.6209	3,974	5,129

$$DPBP = 3 + 2{,}465/3{,}620 = 3 + 0.681 = 3.68 \text{ years} \quad (\approx 3 \text{ years } 8 \text{ months})$$

DPBP = 3.68 years. The investment is recovered in discounted terms before year 4.

SECTION C: Equivalent Worth Methods (PW / FW / AW)

Q5. Define equivalent worth and rate of return methods. How much must you deposit now at 8%/year to draw Rs. 10,000/month for 10 years? [2070 Magh Q3a — 2+4 Marks]

Answer:

Equivalent Worth Methods

Equivalent worth methods convert all cash flows to a single monetary figure at a common reference point — present (PW), future (FW), or annual (AW) — using MARR as the discount rate. If the equivalent worth ≥ 0, the project meets the minimum return requirement.

Rate of Return Method

Finds the interest rate i* (IRR) at which NPW of all cash flows = 0, then compares i* to MARR. Accept if i* ≥ MARR.

Numerical Solution:

$$\text{Monthly rate: } i_m = 8\%/12 = 0.6667\% = 0.006667$$

$$n = 10 \text{ years} \times 12 = 120 \text{ months}, \quad A = \text{Rs. } 10{,}000/\text{month}$$

$$P = A \times (P/A, i_m, 120) = 10{,}000 \times [(1.006667)^{120} - 1] / [0.006667 \times (1.006667)^{120}]$$

$$(1.006667)^{120} = 2.22099$$

$$P = 10{,}000 \times 1.22099 / (0.006667 \times 2.22099) = 10{,}000 \times 1.22099 / 0.014806$$

$$P = 10{,}000 \times 82.462 = \text{Rs. } 8{,}24{,}620$$

Deposit required now: Rs. 8,24,620.

Q6. Calculate CR and make investment decision using AW. I=1,00,000; S=25,000; Revenue=20,000/yr; Expense=5,000/yr; n=10 yrs; MARR=10%. [2081 Ashwin Q3a — 3 Marks]

Answer:

Capital Recovery (CR) Cost:

$$CR = (I - S)(A/P,10\%,10) + S \times i$$

$$(A/P,10\%,10) = 0.16275$$

$$CR = (1,00,000-25,000)(0.16275) + 25,000(0.10) = 12,206 + 2,500 = Rs.\ 14,706$$

Annual Worth:

$$AW = Annual\ Revenue - Annual\ Expense - CR$$

$$AW = 20,000 - 5,000 - 14,706 = Rs.\ 294$$

AW = Rs. 294 > 0: Project marginally meets MARR of 10%. ACCEPTABLE.

Q7. Evaluate the project by AW formulation at i = 12%. CF at t=0,1,2,3,4,5 are -3000, 800, 1000, 1100, 1210, 1464. [2074 Bhadra Q3b — 4 Marks]

Answer:

Step 1: Compute PW at 12%

$$PW = -3,000+800(0.8929)+1,000(0.7972)+1,100(0.7118)+1,210(0.6355)+1,464(0.5674)$$

$$= -3,000+714.3+797.2+783.0+769.0+830.7 = Rs.\ 894.2$$

Step 2: Convert PW to AW

$$AW = PW \times (A/P,12\%,5) = 894.2 \times 0.27741 = Rs.\ 248$$

AW = Rs. 248 > 0: Project earns more than 12%. ACCEPTABLE.

Q8. Calculate both types of B/C ratio using FW formulation. I=5,00,000; Revenue=50,000 yr1 increasing 15,000/yr; Expense=30,000 yr1 decreasing 3%/yr; S=25,000; n=8 yrs; MARR=8%. [2079 Chaitra Q3b, 2079 Ashwin Q3a — 6 Marks]

Answer:

Step 1: FW of Benefits at ε=8%, t=8

$$Revenue:\ A1=50,000,\ G=15,000,\ (F/A,8\%,8)=10.6366,\ (F/G,8\%,8)=[10.6366-8]/0.08=32.96$$

$$FW_rev = 50,000(10.6366)+15,000(32.96)+25,000 = 5,31,830+4,94,400+25,000 = Rs.\ 10,51,230$$

Step 2: FW of Annual Expenses (geometric, g=-3%)

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PW_exp = 30,000×[1-(0.97)^8/(1.08)^8]/0.11 = 30,000×[1-0.7837/1.8509]/0.11
                        = 30,000×0.5766/0.11 = Rs. 1,57,260
FW_exp = 1,57,260×(1.08)^8 = 1,57,260×1.8509 = Rs. 2,91,147
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Step 3: FW of Initial Investment

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FW_inv = 5,00,000×1.8509 = Rs. 9,25,450
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Conventional B/C:

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B/C_conv = FW_benefits / (FW_inv + FW_exp) = 10,51,230/(9,25,450+2,91,147)
                        = 10,51,230/12,16,597 = 0.864
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Modified B/C:

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B/C_mod = (FW_benefits - FW_exp)/FW_inv = (10,51,230-2,91,147)/9,25,450 =
                        7,60,083/9,25,450 = 0.821
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Both B/C < 1.0: Project does NOT meet MARR of 8%. REJECT.

SECTION D: Internal Rate of Return (IRR)

Q9. Compare IRR with NPV. If conflicting, which method to rely on and why? [2082 Kartik Q3b — 3+3 Marks]

Answer:

IRR vs. NPV Comparison:

Basis	IRR	NPV (PW)
Output	Percentage rate (%)	Monetary amount (Rs.)
Decision rule	Accept if IRR ≥ MARR	Accept if NPW ≥ 0
Reinvestment assumption	Reinvestment at IRR	Reinvestment at MARR (realistic)
Multiple solutions	Possible for non-conventional CF	Always unique
Scale of investment	Ignores absolute size	Reflects magnitude of value created
Mutually exclusive ranking	May be incorrect	Always correct

When Conflicting: Rely on NPW / PW Method. Reasons:

1. Realistic Reinvestment: NPW assumes reinvestment at MARR, which is achievable. IRR assumes reinvestment at the (often unrealistically high) IRR.
2. Unique Value: NPW always gives a single unique answer; IRR may give multiple solutions.
3. Accounts for Scale: NPW reflects absolute wealth creation; IRR ignores the size of investment.
4. Correct Ranking: NPW maximises total economic value to the organisation, which is the true investment objective.

Example: Project A invests Rs.10,000, earns Rs.13,000 yr1. IRR=30%. Project B invests Rs.1,00,000, earns Rs.1,25,000 yr1. IRR=25%. At MARR=10%: NPW(A)=Rs.1,818, NPW(B)=Rs.13,636. NPW correctly identifies B as creating 7.5× more value.

In case of conflict for mutually exclusive alternatives: always use NPW as the primary decision criterion.

Q10. Explain drawbacks of IRR with suitable examples. [2079 Ashwin Q3c, 2074 Bhadra Q3a — 4 Marks]

Answer:

Drawback 1: Multiple IRR Values (Non-Conventional Cash Flows)

When cash flows change sign more than once, there can be multiple mathematically valid IRRs, making the decision ambiguous.

Example: EOY 0: -1,000; EOY 1: +4,000; EOY 2: -3,500. Two sign changes yield two IRRs (~17% and ~292%).

Drawback 2: Incorrect Ranking of Mutually Exclusive Alternatives

IRR is a relative percentage measure and may prefer a small high-% project over a large project that creates more total value.

Example: Project X (invest Rs.1,000, earn Rs.1,400 yr1, IRR=40%) vs Project Y (invest Rs.10,000, earn Rs.13,000 yr1, IRR=30%). At MARR=10%: NPW(X)=Rs.273, NPW(Y)=Rs.2,727. IRR prefers X, but Y creates 10× more value.

Drawback 3: Unrealistic Reinvestment Assumption

IRR assumes all intermediate cash flows reinvested at the IRR. A very high IRR (e.g., 35%) implies reinvestment at 35%, which is rarely achievable in practice.

Drawback 4: Does Not Measure Scale of Investment

A project with 50% return on Rs.1,000 creates less absolute value than one with 20% return on Rs.1,00,000.

Drawback 5: Incremental Analysis Required

For mutually exclusive alternatives, IRR requires incremental (marginal) analysis to avoid ranking errors, adding computational complexity.

Solution: Use ERR to overcome drawbacks 1 and 3. Use NPW to overcome drawbacks 2, 4, and 5.

Q11. Explain with example how inconsistent ranking occurs in IRR method. [2079 Chaitra Q4a — 4 Marks]

Answer:

Cause of Inconsistent Ranking:

IRR and NPW give conflicting rankings of mutually exclusive projects when projects differ in (a) scale of investment, (b) useful life, or (c) timing/pattern of cash flows.

Example — Scale Difference (1-year projects, MARR=10%):

	Project A (Small)	Project B (Large)
Initial Investment	Rs. 10,000	Rs. 1,00,000
Cash Inflow at yr1	Rs. 13,000	Rs. 1,22,000
IRR	30%	22%
NPW at 10%	Rs. 1,818	Rs. 10,909

IRR ranks A first (30% > 22%). NPW ranks B first (Rs.10,909 > Rs.1,818). This is the inconsistency.

Why NPW is Correct:

Project B creates Rs.10,909 of value above the 10% minimum; Project A creates only Rs.1,818. The goal is to maximise wealth, not percentage return.

Resolution Using Incremental IRR:

$$\text{Incremental investment (B-A)} = \text{Rs. } 90{,}000$$

$$\text{Incremental cash inflow} = 1{,}22{,}000 - 13{,}000 = \text{Rs. } 1{,}09{,}000$$

$$\text{Incremental IRR} = (1{,}09{,}000/90{,}000) - 1 = 21.1\%$$

Since incremental IRR (21.1%) > MARR (10%), the extra investment in B earns above minimum. Choose B.

Rule: For mutually exclusive alternatives under IRR, always perform incremental IRR analysis, or use NPW to avoid inconsistent rankings.

Q12. Capital investment Rs. 65,000; annual savings at EOY 1-5 are Rs. 25,000; 30,000; 30,000; 40,000; 46,000. Evaluate using DPBP, IRR, AW, ERR. MARR=15%, ε=18%. [2082 Kartik Q4 — 8 Marks]

Answer:

Initial investment = Rs. 65,000. MARR = 15%, ε = 18%

(a) Discounted Payback Period at 15%:

EOY	Net CF (Rs.)	PW @15%	PW of CF	Cumulative PW
0	-65,000	1.000	-65,000	-65,000
1	25,000	0.8696	21,740	-43,260
2	30,000	0.7561	22,683	-20,577
3	30,000	0.6575	19,725	-852
4	40,000	0.5718	22,872	22,020
5	46,000	0.4972	22,871	44,891

$$\text{DPBP} = 3 + 852/22{,}872 = 3.04 \text{ years} \quad (\approx 3 \text{ years})$$

DPBP < 5 years: ACCEPTABLE.

(b) IRR:

$$\text{PW@30\%} = -65{,}000+25{,}000(0.7692)+30{,}000(0.5917)+30{,}000(0.4552)+40{,}000(0.3501)+46{,}000(0.2693)$$

$$= -65{,}000+19{,}230+17{,}751+13{,}656+14{,}004+12{,}388 = +12{,}029$$

$$\text{PW@45\%} = -65{,}000+25{,}000(0.6897)+30{,}000(0.4756)+30{,}000(0.3280)+40{,}000(0.2262)+46{,}000(0.1560)$$

$$= -65{,}000+17{,}241+14{,}268+9{,}839+9{,}049+7{,}176 = -7{,}427$$

$$\text{IRR} = 30+15\times 12{,}029/(12{,}029+7{,}427) = 30+9.27 = 39.27\%$$

IRR = 39.27% >> 15%: ACCEPTABLE.

(c) Annual Worth at 15%:

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PW@15% = -65,000+21,740+22,683+19,725+22,872+22,871 = +44,891
AW = 44,891×(A/P,15%,5) = 44,891×0.2983 = Rs. 13,392
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AW = Rs. 13,392 > 0: ACCEPTABLE.

(d) ERR (ε=18%):

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PW_neg = -65,000  (no other negatives)
FW_pos = 25,000(1.9388)+30,000(1.6430)+30,000(1.3924)+40,000(1.18)+46,000
= 48,470+49,290+41,772+47,200+46,000 = Rs. 2,32,732
65,000(1+ERR)^5 = 2,32,732  =>  (1+ERR)^5 = 3.5805
ERR = 3.5805^(0.2) - 1 = 1.2896 - 1 = 28.96%
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ERR = 28.96% > 15%: ACCEPTABLE. All four methods confirm the project is economically sound.

Q13. Use IRR to evaluate project and make Unrecovered Balance Diagram. CF at t=0,1,2,3,4,5 are -60000, 20000, 40000, -40000, 50000, 70000. MARR=15%. [2077 Chaitra Q4 — 5+1 Marks]

Answer:

Finding IRR:

```
Try i=15%: PW = -60,000+20,000(0.8696)+40,000(0.7561)-
     40,000(0.6575)+50,000(0.5718)+70,000(0.4972)
     = -60,000+17,392+30,244-26,300+28,590+34,804 = +24,730
Try i=35%: PW = -60,000+20,000(0.7407)+40,000(0.5487)-
     40,000(0.4064)+50,000(0.3011)+70,000(0.2230)
     = -60,000+14,814+21,948-16,256+15,055+15,610 = -8,829
IRR = 15+20×24,730/(24,730+8,829) = 15+14.74 = 29.74%
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IRR = 29.74% > MARR (15%): ACCEPTABLE.

Unrecovered Investment Balance (UIB) at IRR = 29.74%:

UIB_end = UIB_beg × (1+IRR) + Net CF

EOY	Net CF (Rs.)	Beg. UIB (Rs.)	Interest @29.74%	End UIB (Rs.)
0	-60,000	—	—	-60,000
1	+20,000	-60,000	-17,844	-57,844
2	+40,000	-57,844	-17,200	-35,044
3	-40,000	-35,044	-10,422	-85,466
4	+50,000	-85,466	-25,419	-60,885
5	+70,000	-60,885	-18,107	≈ 0

The UIB starts at -Rs.60,000 and approaches zero at end of year 5, confirming IRR ≈ 29.74%. This diagram shows how the unrecovered investment reduces over the project life.

Q14. Calculate IRR and show U/B diagram. CF at t=0,1,2,3,4,5 are -40000, -4250, 9280, 38600, 61460, -20000. MARR=12%. [2079 Ashwin Q3b — 6 Marks]

Answer:

Note: Two sign changes (non-conventional CF). Proceed with trial.

Try i=15%: PW = -40,000-4,250(0.8696)+9,280(0.7561)+38,600(0.6575)+61,460(0.5718)-20,000(0.4972)

= -40,000-3,696+7,017+25,379+35,153-9,944 = +13,909

Try i=35%: PW = -40,000-4,250(0.7407)+9,280(0.5487)+38,600(0.4064)+61,460(0.3011)-20,000(0.2230)

= -40,000-3,148+5,092+15,687+18,502-4,460 = -8,327

IRR = 15+20×13,909/(13,909+8,327) = 15+12.5 = 27.5%

IRR = 27.5% > MARR (12%): ACCEPTABLE.

UIB Diagram at 27.5%:

EOY	Net CF (Rs.)	Beg. UIB	Interest @27.5%	End UIB
0	-40,000	—	—	-40,000
1	-4,250	-40,000	-11,000	-55,250
2	+9,280	-55,250	-15,194	-61,164
3	+38,600	-61,164	-16,820	-39,384
4	+61,460	-39,384	-10,831	+11,245
5	-20,000	+11,245	+3,092	≈ 0

UIB approaches zero at end of year 5, confirming IRR ≈ 27.5%.

Q15. I=1,00,000; n=5 yrs; S=0; O&M=25,000/yr; Revenue=60,000 yr1 decreasing 4,000/yr. Draw U/B diagram, evaluate conventional BCR using PW, discounted PBP at MARR=10%. [2070 Bhadra Q3 — 6+5+5 Marks]

Answer:

Net Annual Cash Flows:

EOY	Benefit (Rs.)	O&M (Rs.)	Net CF (Rs.)
1	60,000	25,000	35,000
2	56,000	25,000	31,000
3	52,000	25,000	27,000
4	48,000	25,000	23,000
5	44,000	25,000	19,000

Conventional B/C Ratio (PW formulation) at 10%:

PW_benefit = 60,000(P/A,10%,5) - 4,000(P/G,10%,5)

(P/A,10%,5)=3.7908, (P/G,10%,5)=6.8618

PW_benefit = 60,000(3.7908) - 4,000(6.8618) = 2,27,448 - 27,447 = Rs. 2,00,001

PW_costs = 1,00,000 + 25,000(3.7908) = 1,00,000 + 94,770 = Rs. 1,94,770

B/C_conv = 2,00,001/1,94,770 = 1.027

Conventional B/C = 1.027 > 1.0: ACCEPTABLE.

Discounted Payback Period at 10%:

EOY	Net CF (Rs.)	PW @10%	Disc. CF	Cumulative PW
0	-1,00,000	1.000	-1,00,000	-1,00,000
1	35,000	0.9091	31,819	-68,181
2	31,000	0.8264	25,618	-42,563
3	27,000	0.7513	20,285	-22,278
4	23,000	0.6830	15,709	-6,569
5	19,000	0.6209	11,797	5,228

DPBP = 4 + 6,569/11,797 = 4 + 0.557 = 4.56 years (≈ 4 years 7 months)

DPBP = 4.56 years < project life (5 years): ACCEPTABLE.

UIB / IRR:

Find IRR: At i=12%: PW = -100,000+35,000(0.8929)+31,000(0.7972)+27,000(0.7118)+23,000(0.6355)+19,000(0.5674)

= -100,000+31,252+24,713+19,219+14,617+10,781 = +582

IRR ≈ 12.1%. UIB diagram computed at IRR=12.1% starts at -Rs.1,00,000 and approaches zero at end of year 5.

SECTION E: External Rate of Return (ERR)

Q16. Define IRR. Find IRR and ERR. CF at t=0–6 are -100000, 32000, 48000, 65000, 75000, -38000, 50000. MARR=ε=15%. [2079 Jestha Q3a — 6 Marks]

Answer:

IRR Definition:

IRR (i*) is the interest rate at which NPW of all project cash flows = 0. It represents the project's own earning rate on unrecovered investment. Accept if i* ≥ MARR.

Finding IRR:

Try i=20%: PW = -100,000+32,000(0.8333)+48,000(0.6944)+65,000(0.5787)+75,000(0.4823) -38,000(0.4019)+50,000(0.3349)

= -100,000+26,667+33,333+37,618+36,173-15,272+16,745 = +35,264

Try i=45%: PW = -100,000+32,000(0.6897)+48,000(0.4756)+65,000(0.3280)+75,000(0.2262) -38,000(0.1560)+50,000(0.1076)

= -100,000+22,069+22,829+21,320+16,965-5,928+5,380 = -17,365

IRR = 20+25×35,264/(35,264+17,365) = 20+16.7 = 36.7%

IRR = 36.7% > MARR (15%): ACCEPTABLE.

Finding ERR (ε=15%):

PW_neg = -100,000 + (-38,000)(P/F,15%,5) = -100,000 - 38,000(0.4972) = -118,894

FW_pos = 32,000(F/P,15%,5)+48,000(F/P,15%,4)+65,000(F/P,15%,3)+75,000(F/P,15%,2)+50,000

= 32,000(2.0114)+48,000(1.7490)+65,000(1.5209)+75,000(1.3225)+50,000

= 64,365+83,952+98,859+99,188+50,000 = Rs. 3,96,364

118,894(1+ERR)^6 = 3,96,364 => (1+ERR)^6 = 3.3337

ERR = 3.3337^(1/6) - 1 = 1.2218 - 1 = 22.18%

ERR = 22.18% > MARR (15%): ACCEPTABLE. Note ERR (22.18%) < IRR (36.7%) — ERR is more conservative due to realistic reinvestment at ε=15%.

Q17. Invest Rs. 10,000 at t=0, Rs. 8,000 at t=1. Revenue Rs. 6,000 from year 2 growing 6%/year up to year 10. ε=15%, MARR=20%. Find ERR and assess feasibility. [2079 Chaitra Q4b — 6 Marks]

Answer:

Step 1: PW of negatives at MARR=20%

PW_neg = -10,000 - 8,000(P/F,20%,1) = -10,000 - 8,000(0.8333) = -16,667

Step 2: FW of revenues (geometric, A1=6,000 at t=2, g=6%, ε=15%) at t=10

9 payments from t=2 to t=10. PW at t=1:

PW@t1 = 6,000×[1-(1.06)^9(1.15)^-9]/(0.15-0.06)

(1.06)^9=1.6895, (1.15)^9=3.5179

= 6,000×[1-1.6895/3.5179]/0.09 = 6,000×0.5198/0.09 = Rs. 34,654

FW_pos = 34,654×(1.15)^9 = 34,654×3.5179 = Rs. 1,21,909

Step 3: Solve for ERR

16,667(1+ERR)^10 = 1,21,909

(1+ERR)^10 = 1,21,909/16,667 = 7.315

ERR = 7.315^(0.1) - 1 = 1.220 - 1 = 22.0%

ERR = 22.0% > MARR (20%): Project is ACCEPTABLE. The margin is small (2%) but sufficient.

Q18. Compute ERR for project with CF at t=0–6 of -300000, 150000, 200000, -100000, 200000, 150000, -50000. MARR=12%, ε=15%. [2076 Baisakh Q3a — 4 Marks]

Answer:

Step 1: PW of negatives at MARR=12%

```
PW_neg = -3,00,000 + (-1,00,000)(P/F,12%,3) + (-50,000)(P/F,12%,6)
       = -3,00,000 - 1,00,000(0.7118) - 50,000(0.5066)
       = -3,00,000 - 71,180 - 25,330 = -3,96,510
```

Step 2: FW of positives at ε=15%, t=6

```
FW_pos =
1,50,000(F/P,15%,5)+2,00,000(F/P,15%,4)+2,00,000(F/P,15%,2)+1,50,000(F/P,15%,1)
= 1,50,000(2.0114)+2,00,000(1.7490)+2,00,000(1.3225)+1,50,000(1.15)
= 3,01,710+3,49,800+2,64,500+1,72,500 = Rs. 10,88,510
```

Step 3: Solve for ERR

```
3,96,510(1+ERR)^6 = 10,88,510
(1+ERR)^6 = 2.7451  =>  ERR = 2.7451^(1/6) - 1 = 1.1833 - 1 = 18.33%
```

ERR = 18.33% > MARR (12%): ACCEPTABLE.

SECTION F: Benefit-Cost (B/C) Ratio Method

Q19. Recommend the best project from three mutually exclusive projects using modified B/C ratio. MARR=15%. [2081 Chaitra Q3b, 2080 Chaitra Q3b — 6 Marks]

Answer:

	Project A	Project B	Project C
Capital Investment (Rs.)	40,000	60,000	80,000
Annual Revenues (Rs.)	3,000	3,500	4,000
Annual Expenses (Rs.)	300	400	500
Useful Life (years)	5	5	5
Salvage Value (Rs.)	400	700	900

MARR=15%, (A/P,15%,5)=0.29832

Capital Recovery (CR) for each project:

```
CR_A = (40,000-400)(0.29832)+400(0.15) = 11,813+60 = Rs. 11,873
CR_B = (60,000-700)(0.29832)+700(0.15) = 17,700+105 = Rs. 17,805
CR_C = (80,000-900)(0.29832)+900(0.15) = 23,597+135 = Rs. 23,732
```

Modified B/C = (AW Revenue - AW Expense) / CR:

```
Modified B/C_A = (3,000-300)/11,873 = 2,700/11,873 = 0.227
Modified B/C_B = (3,500-400)/17,805 = 3,100/17,805 = 0.174
Modified B/C_C = (4,000-500)/23,732 = 3,500/23,732 = 0.147
```

All three projects have Modified B/C < 1.0. None is economically viable at MARR=15%. The 'Do Nothing' option is best. If forced to choose, Project A has the least-poor ratio (0.227).

Q20. Machine costs Rs. 4,00,000; S=1,00,000; n=5 yrs; Revenue increase=1,50,000/yr; Fuel cost=30,000/yr. Find IRR and B/C when MARR=10%. [2081 Ashwin Q3b — 3+3 Marks]

Answer:

Net Annual Benefit = 1,50,000 - 30,000 = Rs. 1,20,000

(A) IRR:

```
Try i=18%: PW = -4,00,000+1,20,000(3.1272)+1,00,000(0.4371) = -
               400,000+375,264+43,710 = +18,974
Try i=25%: PW = -4,00,000+1,20,000(2.6893)+1,00,000(0.3277) = -
               400,000+322,716+32,770 = -44,514
IRR = 18+7×18,974/(18,974+44,514) = 18+7×0.299 = 18+2.09 = 20.09%
```

IRR = 20.09% > MARR (10%): ACCEPTABLE.

(B) B/C Ratio at MARR=10%:

```
CR = (4,00,000-1,00,000)(A/P,10%,5)+1,00,000(0.10)
= 3,00,000(0.26380)+10,000 = 79,140+10,000 = Rs. 89,140
Conventional B/C = 1,50,000/(89,140+30,000) = 1,50,000/1,19,140 = 1.259
Modified B/C = (1,50,000-30,000)/89,140 = 1,20,000/89,140 = 1.346
```

Both B/C > 1.0: ACCEPTABLE.

Q21. I=2,50,000; Revenue=75,000 yr1 increasing 5,000/yr; O&M=15,000/yr; S=25,000; MARR=10%; n=6 yrs. Find both types of B/C using FW formulation. [2079 Chaitra Q3b — 6 Marks]

Answer:

Step 1: FW of Benefits (Revenues + Salvage):

```
(F/A,10%,6)=7.7156,  (F/G,10%,6)=[(7.7156-6)]/0.10=17.156
FW_rev = 75,000(7.7156)+5,000(17.156)+25,000 = 5,78,670+85,780+25,000 =
                                    Rs. 6,89,450
```

Step 2: FW of Costs:

```
FW_inv = 2,50,000×(1.10)^6 = 2,50,000×1.7716 = Rs. 4,42,900
FW_OM  = 15,000×7.7156 = Rs. 1,15,734
```

Step 3: B/C Ratios:

```
Conventional B/C = 6,89,450/(4,42,900+1,15,734) = 6,89,450/5,58,634 =
                                    1.234
Modified B/C = (6,89,450-1,15,734)/4,42,900 = 5,73,716/4,42,900 = 1.295
```

Both B/C > 1.0: Project is ACCEPTABLE at MARR=10%.

SECTION G: Comprehensive Multi-Method Problems

Q22. I=5,00,000; n=8 yrs; Revenue=50,000 yr1 increasing 15,000/yr; Expense=30,000 yr1 decreasing 3%/yr; S=25,000; MARR=8%. Decide using PW, IRR, B/C (both), DPBP. [2080 Ashwin Q3a — 3+3+2+2 Marks]

Answer:

PW of Revenues (arithmetic gradient, i=8%, n=8):

$$PW_rev = 50{,}000(P/A,8\%,8)+15{,}000(P/G,8\%,8)$$

$$(P/A,8\%,8)=5.7466, \quad (P/G,8\%,8)=17.806$$

$$PW_rev = 50{,}000(5.7466)+15{,}000(17.806) = 2{,}87{,}330+2{,}67{,}090 = Rs.\ 5{,}54{,}420$$

PW of Expenses (geometric, g=-3%, i=8%, n=8):

$$PW_exp = 30{,}000\times[1-(0.97)^8/(1.08)^8]/0.11 = 30{,}000\times[1-0.4234]/0.11 = Rs.\ 1{,}57{,}260$$

PW of Salvage:

$$PW_S = 25{,}000(P/F,8\%,8) = 25{,}000(0.5403) = Rs.\ 13{,}508$$

(a) Net PW:

$$NPW = -5{,}00{,}000+5{,}54{,}420-1{,}57{,}260+13{,}508 = -Rs.\ 89{,}332$$

NPW < 0: Project does NOT meet MARR=8%. REJECT.

(b) IRR:

Since NPW < 0 at 8%, IRR < 8% < MARR. Project fails IRR criterion. REJECT.

(c) B/C Ratios:

$$\text{Conventional } B/C = (5{,}54{,}420+13{,}508)/(5{,}00{,}000+1{,}57{,}260) = 5{,}67{,}928/6{,}57{,}260 = 0.864 < 1.0$$

$$\text{Modified } B/C = (5{,}67{,}928-1{,}57{,}260)/5{,}00{,}000 = 4{,}10{,}668/5{,}00{,}000 = 0.821 < 1.0$$

Both B/C < 1.0: REJECT.

(d) Discounted Payback Period:

Since NPW is negative throughout, the project never recovers its investment in discounted terms. DPBP > project life. REJECT.

Conclusion: All four methods unanimously reject this project at MARR=8%. Do not invest.

Q23. I=4,00,000; n=8 yrs; S=1,00,000; Revenue=1,60,000 yr1 decreasing 10,000/yr; Expenses=40,000 yr1 increasing 5,000/yr; MARR=9%. Decide using IRR, B/C, DPBP. [2075 Bhadra Q3a — 4+4+4 Marks]

Answer:

Net Annual Cash Flows:

EOY	Revenue	Expense	Net CF
1	1,60,000	40,000	1,20,000
2	1,50,000	45,000	1,05,000

3	1,40,000	50,000	90,000
4	1,30,000	55,000	75,000
5	1,20,000	60,000	60,000
6	1,10,000	65,000	45,000
7	1,00,000	70,000	30,000
8	90,000+1,00,000	75,000	1,15,000

(i) IRR:

Try i=10%: PW = -4,00,000+1,20,000(0.9091)+1,05,000(0.8264)+90,000(0.7513)+75,000(0.6830) + 60,000(0.6209)+45,000(0.5645)+30,000(0.5132)+1,15,000(0.4665) = -4,00,000+109,092+86,772+67,617+51,225+37,254+25,403+15,396+53,648 = +46,407

Try i=18%: PW = -4,00,000+1,20,000(0.8475)+1,05,000(0.7182)+90,000(0.6086)+75,000(0.5158) + 60,000(0.4371)+45,000(0.3704)+30,000(0.3139)+1,15,000(0.2660) = -4,00,000+101,700+75,411+54,774+38,685+26,226+16,668+9,417+30,590 = -46,529

IRR = 10+8×46,407/(46,407+46,529) = 10+4.0 = 14.0%

IRR = 14% > MARR (9%): ACCEPTABLE.

(ii) B/C Ratio at MARR=9%:

PW_gross_rev = 1,60,000(P/A,9%,8) - 10,000(P/G,9%,8)

(P/A,9%,8)=5.5348, (P/G,9%,8)=14.973

PW_gross_rev = 1,60,000(5.5348)-10,000(14.973) = 8,85,568-1,49,730 = 7,35,838

PW_S = 1,00,000(P/F,9%,8) = 1,00,000(0.5019) = 50,190

Total PW_benefit = 7,35,838+50,190 = Rs. 7,86,028

PW_exp = 40,000(5.5348)+5,000(14.973) = 2,21,392+74,865 = Rs. 2,96,257

Conventional B/C = 7,86,028/(4,00,000+2,96,257) = 7,86,028/6,96,257 = 1.129

Modified B/C = (7,86,028-2,96,257)/4,00,000 = 4,89,771/4,00,000 = 1.224

Both B/C > 1.0: ACCEPTABLE.

(iii) Discounted Payback Period at 9%:

EOY	Net CF	PW @9%	Disc. CF	Cumulative PW
0	-4,00,000	1.000	-4,00,000	-4,00,000
1	1,20,000	0.9174	1,10,088	-2,89,912
2	1,05,000	0.8417	88,379	-2,01,533
3	90,000	0.7722	69,498	-1,32,035
4	75,000	0.7084	53,130	-78,905

5	60,000	0.6499	38,994	-39,911
6	45,000	0.5963	26,834	-13,077
7	30,000	0.5470	16,410	+3,333

DPBP = 6+13,077/16,410 = 6.80 years (≈ 6 years 10 months)

DPBP = 6.8 years < project life (8 years): ACCEPTABLE. All three methods confirm the project is economically viable.

SECTION H: Financial vs. Economic Analysis & Life Cycle Costing

Q24. What do you mean by financial and economic analysis? Briefly explain the concept of lifecycle costing. [2075 Bhadra Q3b, 2078 Chaitra Q3c, 2079 Ashwin Q3c — 2+2 or 4 Marks]

Answer:

Financial Analysis

Financial analysis (private sector / commercial analysis) evaluates a project from the perspective of the private investor or organisation. It uses actual market prices and focuses on whether the investment will provide an acceptable monetary return to the investor.

- Viewpoint: Private organisation / investor.
- Goal: Profit maximisation; achieving required rate of return (MARR / hurdle rate).
- Prices used: Actual market prices.
- Taxes: Treated as real costs (reduce profit).
- Metrics: NPW, IRR, AW, Payback Period, B/C.
- Scope: Only direct cash flows to the organisation (revenue, costs, taxes).
- Example: A private company deciding whether to purchase new equipment.

Economic Analysis

Economic analysis (public sector / social cost-benefit analysis) evaluates a project from the perspective of society as a whole. It captures all economic impacts including externalities, social costs, and public benefits that are not captured by market prices.

- Viewpoint: Society / Government / Nation.
- Goal: Maximise net social welfare; efficient use of public resources.
- Prices used: Economic (shadow) prices reflecting true social value.
- Taxes: Treated as transfer payments (not real costs to society; ignored).
- Metrics: Benefit-Cost Ratio (B/C), Economic IRR (EIRR), Social NPV.
- Scope: All direct and indirect effects — environmental impacts, employment, accessibility, health, etc.
- Example: Government deciding whether to build a national highway.

Basis	Financial Analysis	Economic Analysis
Perspective	Private investor	Society / Government
Objective	Profit / ROI	Social welfare / net benefit
Prices	Market prices	Shadow / economic prices
Tax	Taxes are costs	Taxes are transfers (excluded)

Externalities	Excluded	Included
Main Metric	NPW, IRR, AW	B/C Ratio, EIRR

Life Cycle Costing (LCC)

Life Cycle Costing (LCC) is a systematic method for calculating the total cost of owning and using an asset over its entire life — from conception through design, construction, operation, maintenance, and final disposal (cradle-to-grave analysis).

```
LCC = Capital Cost + PW(O&M Costs) + PW(Renewal Costs) + PW(Disposal) -
                                PW(Salvage Value)
```

Phases of Life Cycle and Associated Costs:

1. Concept / Feasibility: Preliminary design and feasibility study costs.
2. Design: Detailed engineering design, specification, and modelling costs.
3. Construction / Acquisition: Initial capital investment — materials, labour, equipment.
4. Operation: Ongoing running costs — energy, labour, consumables, insurance.
5. Maintenance: Preventive maintenance, repairs, periodic overhauls.
6. Disposal / Decommissioning: End-of-life removal, cleanup, salvage recovery.

Importance of LCC: In infrastructure projects, O&M costs over a 30–50 year life often dwarf the initial capital cost. A building with lower construction cost but higher energy/maintenance costs may have a much higher LCC than a more expensive but energy-efficient building. LCC analysis ensures the true long-term economic cost is considered, not just the upfront investment.

Example: Bridge A costs Rs.50 Cr to build, Rs.0.5 Cr/year to maintain; Bridge B costs Rs.65 Cr to build, Rs.0.1 Cr/year to maintain. Over 40 years at 10% MARR, LCC analysis reveals which is truly more economical by comparing the PW of all lifetime costs.

QUICK REFERENCE SUMMARY — Chapter 3 Formulas

Method	Accept Condition	Formula
PW (NPW)	NPW ≥ 0	NPW = Σ[NCF_t × (P/F,i,t)]
AW	AW ≥ 0	AW = NPW × (A/P,i,n)
FW	FW ≥ 0	FW = NPW × (F/P,i,n)
IRR	IRR ≥ MARR	NPW(i*) = 0; solve for i*
ERR	ERR ≥ MARR	PW_neg×(F/P,ERR,n) = FW_pos
Conventional B/C	≥ 1.0	PW(Benefits) / PW(Investment+O&M)
Modified B/C	≥ 1.0	(PW_Benefits - PW_O&M) / PW_Investment
Discounted PBP	< Project Life	Cumulative PW of NCF = 0
Standard PBP	< Cutoff	I / Annual NCF (uniform only)

Capital Recovery: CR = (I - S)(A/P,i,n) + S×i

UIB (End of period) = UIB_prev × (1 + IRR) + NCF_t [UIB at t=n ≈ 0 confirms IRR]

—— End of Chapter 3 Questions & Answers ——